Análisis del circuito por superposición

Circuito 1. Apagando $F(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

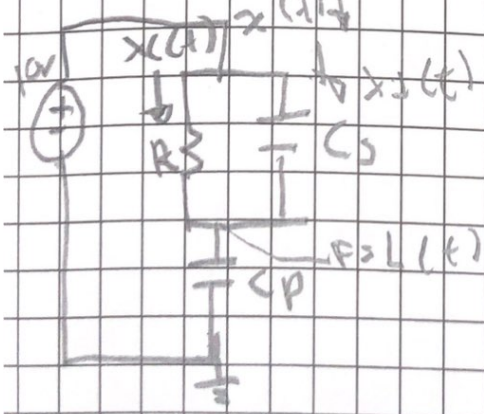
$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

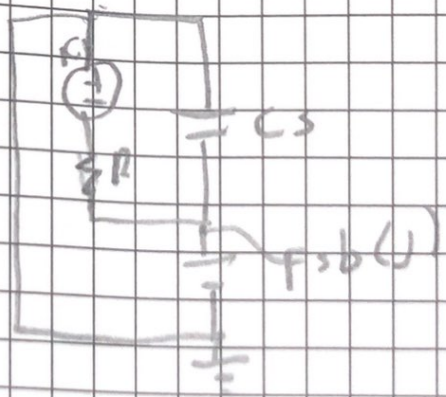
$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

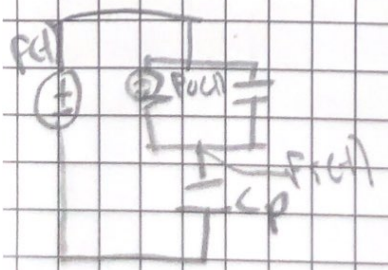
$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p F_{s1}(s) = C_s F(s) - C_s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$\left(C_p + C_s + \frac{1}{Rs}\right) F_{s1}(s) = \left(C_s + \frac{1}{Rs}\right) F(s)$$







$$\frac{R(C_p+C_s)s+1}{R}, \quad F_{s1}(s) = \frac{C_s R s+1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s+1}{R} F(s)}{\frac{R(C_p+C_s)s+1}{R}} = \frac{C_s R s+1}{R(C_p+C_s)s+1} F(s)$$

Circuito 2: Apagando $f(t)$

$$C_p \frac{d[f_{s2}(s)]}{dt} = C_s \frac{d[x(t)-f_{s2}(t)]}{dt} + \frac{x(t)-f_{s2}(t)}{R}$$

$$C_p s F_{s2}(s) = C_s s[X(s)-F_{s2}(s)] + \frac{X(s)-F_{s2}(s)}{R}$$

$$C_p s F_{s2}(s) = C_s s X(s) - C_s s F_{s2}(s) + \frac{X(s)}{R} - \frac{F_{s2}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s2}(s) = \left(C_s s + \frac{1}{R}\right) X(s)$$

$$\frac{F_{s2}(s)}{X(s)} = \frac{C_s s + \frac{1}{R}}{C_p s + C_s s + \frac{1}{R}}$$

$$\frac{F_{s2}(s)}{X(s)} = \frac{R C_s s+1}{R(C_p+C_s)s+1}$$

$$-\alpha F(t) = R\,x(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$-\alpha F(s) = R\,X(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{X(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{-\frac{X(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1}F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$